Dr. Derma: Leveraging Deep Learning for Reliable Skin Lesion Classification

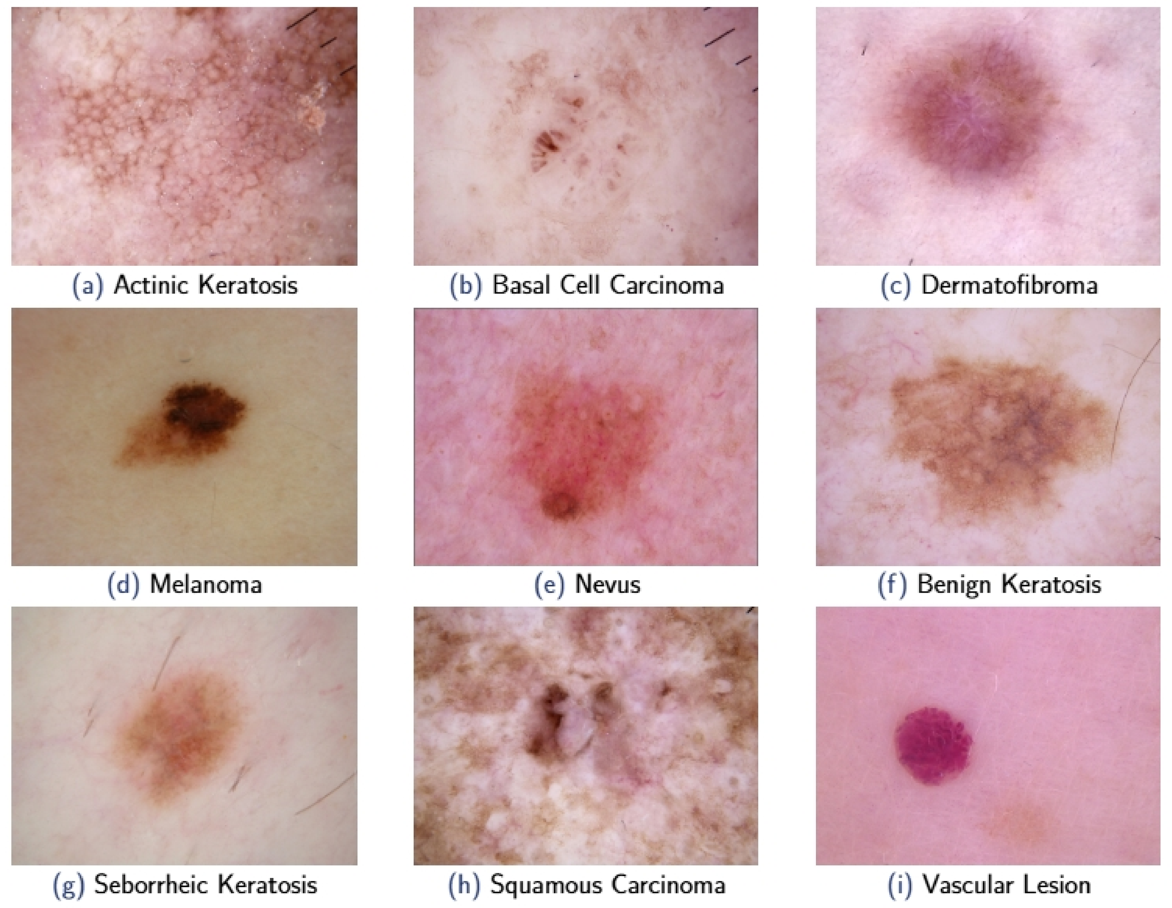
Samajah Abburi Sanskriti Aripineni Aarya Desai Hamza Khawaja
Project Lead: Rushi Trivedi Faculty Advisor: Dr. Kim Dohyeong

Introduction

Skin cancer is one of the most common types of cancer worldwide. Traditional diagnostic methods often rely on the size and appearance of lesions, which can vary greatly, leading to unreliable classifications. Additionally, there is a global shortage of dermatologists, resulting in a lack of timely medical attention. Our project, Dr. Derma, aims to address these challenges by leveraging advanced deep learning technologies. We focus on improving the accuracy and reliability of automated skin cancer detection through the use of data augmentation techniques. This data is trained on a convolutional neural network (CNN) based on the VGG architecture to perform multi-class classification of skin lesions. This approach enhances the model's ability to generalize across diverse image conditions, simulating real-world scenarios and reducing overfitting to specific attributes.

Data Augmentation of the Dataset

- **The International Skin Imaging Collaboration (ISIC) Dataset:** It contains 2,357 images: 2,213 for training and 144 for testing, organized into 9 subfolders, each representing a type of skin lesion.
- **Human Against Machine 10,000 (HAM10000) Dataset:** It contains 10,000 images representing 7 types of skin lesions. We split the dataset into 70% for training, 15% for validation, and 15% for testing.



(a) Actinic Keratosis (b) Basal Cell Carcinoma (c) Dermatofibroma
(d) Melanoma (e) Nevus (f) Benign Keratosis
(g) Seborrheic Keratosis (h) Squamous Carcinoma (i) Vascular Lesion

Figure 1. Types of Skin Lesions

- **Data Augmentation:** It is a technique that diversifies the dataset by creating modified versions of existing images, which helps enhance model accuracy and reduce over-fitting. We applied three augmentation techniques: rotation, contrast adjustments, and zoom.

Model

After receiving the augmented images, the model normalizes them to 224x224 pixels and feeds them into the input layer. Using the VGG19 architecture, the images pass through multiple Conv2D layers that progressively extract features such as edges, textures, and patterns, with deeper layers identifying more complex structures like curved shapes or irregular textures.

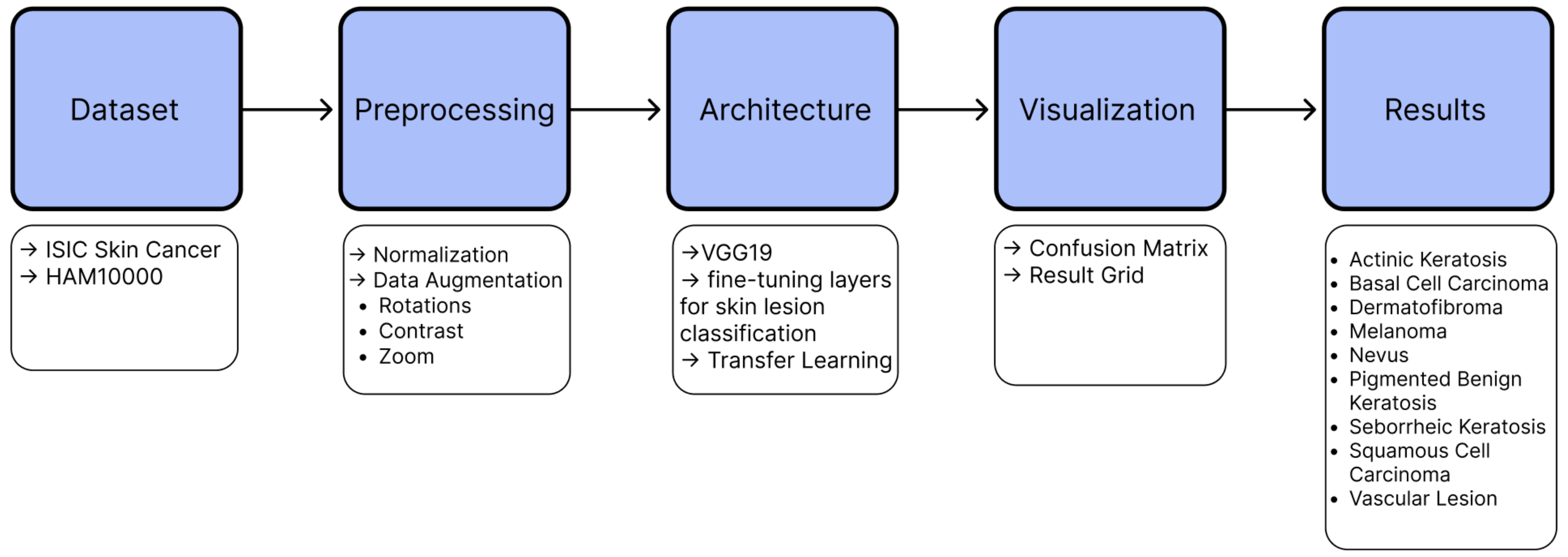

Dataset
Preprocessing
Architecture
Visualization
Results
→ ISIC Skin Cancer
→ HAM10000
→ Normalization
→ Data Augmentation
• Rotations
• Contrast
• Zoom
→VGG19
→ fine-tuning layers for skin lesion classification
→ Transfer Learning
→ Confusion Matrix
→ Result Grid
• Actinic Keratosis
• Basal Cell Carcinoma
• Dermatofibroma
• Melanoma
• Nevus
• Pigmented Benign Keratosis
• Seborrheic Keratosis
• Squamous Cell Carcinoma
• Vascular Lesion

Figure 2. Model Architecture

ReLU activation is applied after each convolution to introduce non-linearity; this enables the model to learn complex patterns, while MaxPooling operations are used to reduce spatial dimensions and retain the most critical features. After the final MaxPooling operation, the flatten layer takes the 3D feature maps and flattens them into a 1D vector, which is then passed through dense layers that learn relationships between the features. The final dense layer, with a softmax activation function, outputs class probabilities, and the predicted labels are compared to the ground truth to evaluate performance on the test set.

Analysis

Dataset Evolution: Initial approach used **ISIC-9** dataset alone with nine distinct lesion types. To address significant class imbalance issues, particularly in underrepresented lesion categories, we combined **ISIC-9** with **HAM10000**. This merger expanded our training dataset to **15,298** images, providing better representation across all classes.

$$\text{Accuracy} = \frac{\text{Correct}}{\text{Total}} \times 100\% \qquad \text{Loss} = -\frac{1}{N}\sum y_i \log(\hat{y}_i) \qquad \text{F1} = 2\frac{\text{P} \times \text{R}}{\text{P} + \text{R}}$$

Dataset	Test Accuracy	Loss	Macro F1
ISIC-9	51.56%	1.7688	77.16%
ISIC-9 + HAM10000	73.75%	1.160	62.6%

Key Improvement: The combined dataset approach led to a significant **22.19%** increase in test accuracy while reducing model loss by more than half.

Results

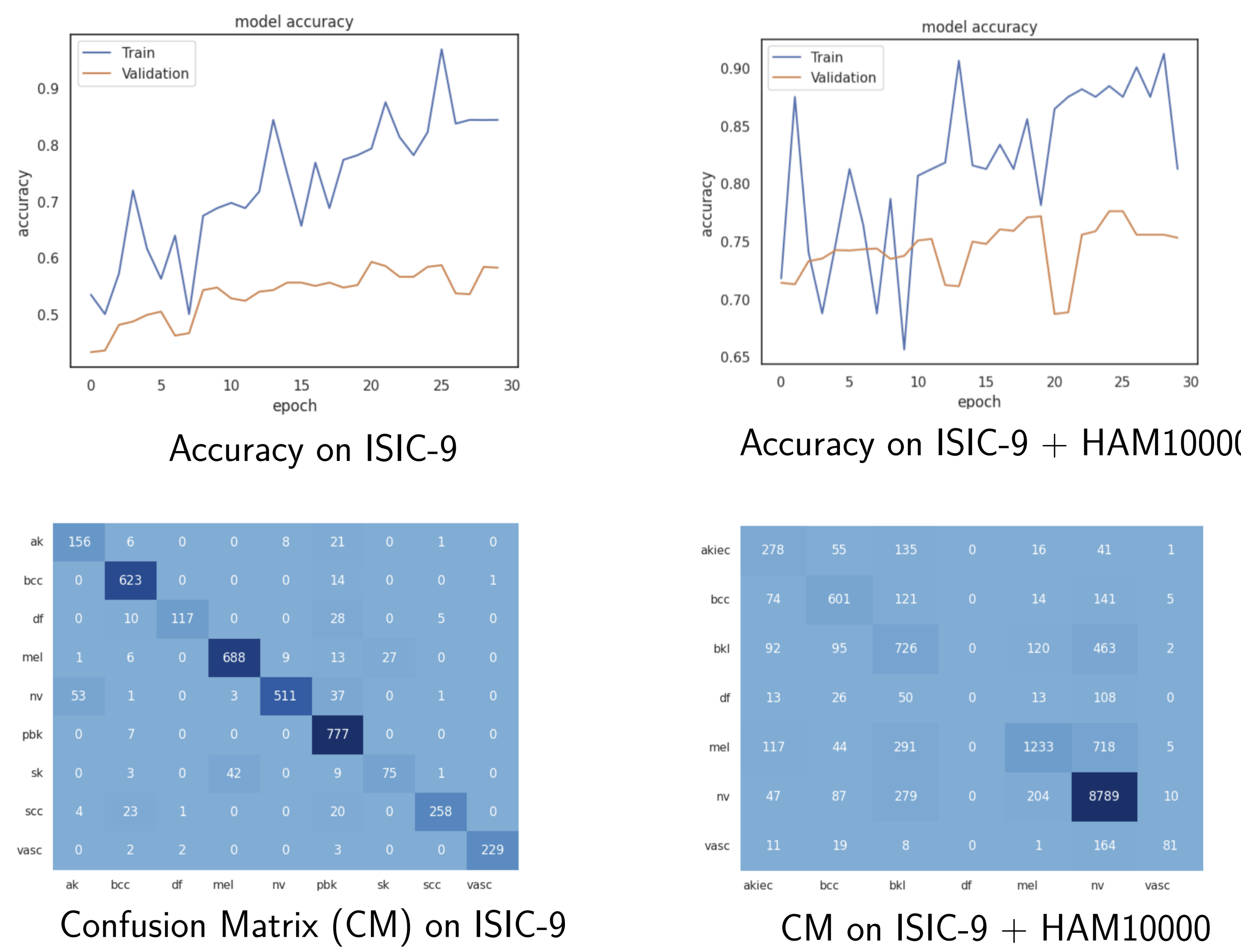

model accuracy
Train
Validation
accuracy
epoch
Accuracy on ISIC-9
Accuracy on ISIC-9 + HAM10000
ak bcc df mel nv pbk sk scc vasc
akiec bcc bkl df mel nv vasc
Confusion Matrix (CM) on ISIC-9
CM on ISIC-9 + HAM10000

Figure 3. Accuracy of ISIC and HAM10000 Datasets

Conclusion

Our model, Dr. Derma, has achieved state-of-the-art results in skin cancer detection, leveraging the powerful VGG19 architecture alongside advanced data augmentation techniques for multi-class classification. Trained on a combination of two datasets, Dr. Derma demonstrates the importance of diverse and extensive data in enhancing both model accuracy and robustness. This cutting-edge model establishes a promising foundation for an accessible diagnostic tool, supporting healthcare professionals in making timely and precise skin cancer diagnoses.

References

- Akter, Mst Shapna, et al. "Multi-class skin cancer classification architecture based on Deep Convolutional Neural Network." 2022 IEEE International Conference on Big Data (Big Data), 17 Dec. 2022, pp. 5404–5413, https://doi.org/10.1109/bigdata55660.2022.10020302.
- DiSanto, Nick, et al. "Leveraging Contextual Data Augmentation for Generalizable Melanoma Detection." arXiv.Org, 9 Aug. 2023, arxiv.org/abs/2212.05116.

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